**East West University**
**Department of Computer Science and Engineering**
**Course: CSE302**
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# LAB 01: Introduction to Oracle/MySQL Database Systems, Simple DDL and DML Queries

**Objectives:**

- Understand and apply basic **DDL** and **DML** operations in SQL.
- Write **SELECT queries** to retrieve and filter data from tables.
- Gain hands-on experience with common **query patterns** used in database applications.

**Topics Covered**

- DDL (Data Definition Language):
    - Creating tables
    - Understanding table structure
- DML (Data Manipulation Language):
    - Inserting data
    - Querying data using `SELECT`
    - Filtering with `WHERE`, `BETWEEN`, `AND`
    - Using comparison operators and logical expressions
    - Retrieving all or selected columns
    - Working with textual and numeric data

**Lab Outcomes**

After completing this lab, students will be able to:

1. Understand basic **DDL operations** such as creating tables (CREATE TABLE).
2. Use **DML commands** like `INSERT` and `SELECT` to manipulate and access data.
3. Write queries to:
    - Retrieve specific columns
    - Apply conditions with `WHERE`, `BETWEEN`, `AND`.
    - Filter data using numeric and text-based criteria
4. Retrieve complete table data with `SELECT *`.
5. Practice combining **logical and comparison operators** for more complex queries.

6. Develop confidence in writing **basic, real-world SQL queries**.

## Lab Activities

### A. SQL Commands Overview: DDL

#### 1. SQL CREATE DATABASE Statement

Used to create a new database.

**Syntax:**

```
CREATE DATABASE database_name;
```

#### 2. SQL CREATE TABLE Statement

Used to create a new table in a database.

**Syntax:**

```
CREATE TABLE table_name (
    column1 datatype,
    column2 datatype,
    column3 datatype,
    ...
);
```

Basic Rules for Creating Tables in MySQL:

- Table names can contain letters, numbers, and underscores (a–z, 0–9, _).

- Avoid using MySQL reserved words (e.g., SELECT, CREATE).

#### 3. ALTER Statement

Used to modify the structure of an existing table.

**Syntax:**

a. ALTER TABLE table_name

ADD column_name datatype;

b. ALTER TABLE table_name

MODIFY column_name new_datatype;

c. ALTER TABLE table_name

DROP COLUMN column_name;

d. ALTER TABLE table_name

CHANGE old_column_name new_column_name datatype;

**4. DROP Statement**

Used to permanently delete a table or a database.

**Syntax:**

DROP TABLE table_name;

DROP DATABASE database_name;

**5. TRUNCATE Statement**

Used to remove all rows from a table but keep the structure.

**Syntax:**

TRUNCATE TABLE table_name;

**6. RENAME Statement**

Used to rename a table.

**Syntax:**

RENAME TABLE old_name TO new_name;

## B. SQL Commands Overview: DML

DML (Data Manipulation Language)

DML is used to manage and manipulate the data within existing tables.

Common DML Commands:

- INSERT : Add new rows of data
- SELECT : Retrieve data
- UPDATE : Modify existing data
- DELETE : Remove data from table

**1. INSERT – Add New Data**

**Syntax:**

```
INSERT INTO table_name (column1, column2, ...)
VALUES (value1, value2, ...);
```

Example:

```
INSERT INTO students (id, name, age)
VALUES (1, 'Ayon', 20);
```

**2. SELECT – Retrieve Data**

**Syntax:**

```
SELECT column1, column2
FROM table_name
WHERE condition;
```

Example:

SELECT name, age FROM students

WHERE age > 18;

**3. UPDATE – Modify Existing Data**

**Syntax:**

UPDATE table_name

SET column1 = value1

WHERE condition;

Example:

UPDATE students

SET age = 21

WHERE id = 1;

**4. DELETE – Remove Data**

**Syntax:**

DELETE FROM table_name

WHERE condition;

Example:

DELETE FROM students

WHERE id = 1;

## Problem Practice:

**1.(a).** Write SQL statement to create a table 'instructor_your_student_id' which has 4 attributes –

i) id (int)
ii) name (varchar)
iii) dept_name (varchar)
iv) salary (decimal)
Example: create table instructor_2020360001 ( …… );

**1.(b).** Write SQL statement to create a table 'course_your_student_id' which has 4 attributes –

i) course_id (varchar)
ii) title (varchar)
iii) dept_name (varchar)
iv) credits (int)
Example: create table course_2020360001 ( …… );

**Lab Task # 02 (Inserting data into a table):**

**2.(a).** Write SQL statements to insert the following records into 'instructor_your_student_id' table:

| ID | Name | Dept_Name | Salary |
|---|---|---|---|
| 10101 | Srinivasan | Comp. Sci. | 65000.50 |
| 12121 | Wasif | Finance | 90000 |
| 15151 | Mozart | Music | 40000 |
| 22222 | Einstein | Physics | 95000 |
| 32343 | El Said | History | 60000 |
| 33456 | Goblin | Physics | 87000.50 |
| 45565 | Katz | Comp. Sci. | 75000 |
| 58583 | Califieri | History | 62000 |

**2.(b).** Write SQL statements to insert the following records into 'course_your_student_id' table:

| course_id | title | dept_name | credits |
|---|---|---|---|
| BIO-101 | Intro. to Biology | Biology | 4 |
| BIO-301 | Genetics | Biology | 4 |
| BIO-399 | Computational Biology | Biology | 3 |
| CS-101 | Intro. to Computer Science | Comp. Sci. | 4 |
| CS-190 | Game Design | Comp. Sci. | 4 |
| CS-319 | Image Processing | Comp. Sci. | 3 |
| CS-347 | Database System Concepts | Comp. Sci. | 3 |
| EE-181 | Intro. to Digital Systems | Elec. Eng. | 3 |
| FIN-201 | Investment Banking | Finance | 3 |

**Lab Task #03: Writing SQL Queries**

i. Show the names of all instructors.
ii. Show the course ID and title of all courses.
iii. Find the name and department of the instructor whose ID is 22222.
iv. Find the titles and credits of courses offered by the 'Comp. Sci.' department.
v. Find the names and departments of instructors who have a salary greater than 70,000.
vi. Find the titles of courses that have credits not less than 4.
vii. Find the names and departments of instructors who have a salary between 80,000 and 100,000 (inclusive).
viii. Find the titles and credits of courses not offered by the 'Comp. Sci.' department.
ix. Display all records from the instructor table.
x. Find all courses (show all columns) that are offered by the 'Biology' department and have credits not equal to 4.
xi. Show unique department names only from the course table.

**Submission Instructions:**

1. For each and every question:
    - Take **screenshots** of the **SQL query execution** and **output/result** in XAMPP or your preferred database environment.
    - Insert both the **SQL solution** and the corresponding **output image** into a **Word document (.doc/.docx)**.
2. **Save the document as a PDF** using the following file naming format: 2022-1-60-001_LAB01.pdf
   *(Replace the ID with your own student ID.)*
3. **Submit** the PDF file using the designated assignment section provided in the Classroom.

**NOTE:**
Before running any SQL queries in your database environment, always **copy and paste** them into **Notepad**.